Session Progress — 2026-04-27

C. elegans Locomotion Model

April 27, 2026

Contents

Overview

This session focused on building up oscillatory circuit complexity from a single neuron toward a single-segment network test, using a slow recovery variable w as the primary mechanism for plateau termination.

1 Single-Neuron Recovery Variable

Objective: Confirm that the recovery variable w reliably returns the neuron to rest after a plateau, even for weak (barely-threshold) stimuli.

Validated Parameters

- $\beta = 0.2$ nS/mV — slope of $w_\infty(V) = \beta \cdot \max(V - T, 0)$ above threshold
- $\tau_w = 200$ ms — recovery timescale

Key Finding

Recovery is always mathematically guaranteed once the neuron reaches the plateau (H), because $f(H) = 0$ and the $-w/C$ term always pulls V downward. The bottleneck is speed, not recoverability.

There is an unstable fixed point in the (V, w) phase plane at:

$$V^* = H + \frac{\beta}{g \cdot m_3} = -41 \text{ mV} \tag{1}$$

$$w^* = \beta\,(V^* - T) = 0.8 \text{ pA} \tag{2}$$

The neuron returns to rest once w accumulates past $w^* \approx 0.8$ pA, which takes approximately 108 ms from $w = 0$ with these parameters.

Test file: `tests/test_single_neuron.py`

2 Mutual Inhibition Circuit (II)

Objective: Test whether w alone can drive oscillations in a two-cell mutually inhibitory circuit.

Result: Failed. $w_\infty(H) = \beta\,(H - T) = 2.0$ pA saturates at or below the constant I_{app} values tested (2–4 pA). w can never overcome a sustained constant drive, so neurons lock permanently to the plateau.

Conclusion: Synaptic fatigue (SF) remains necessary in the II circuit. SF was restored to `tests/test_two_II.py`. The failure is specific to constant I_{app} — in the full network, neurons receive transient synaptic input rather than a sustained DC drive, so the constraint is less severe.

3 Excitatory-Inhibitory Circuit (EI)

Objective: Test whether w alone (no SF) can generate intrinsic oscillations in a two-cell EI circuit.

Circuit: N_0 (excitatory) $\rightarrow$ N_1 (inhibitory) $\rightarrow$ N_0. Constant I_{app} on N_0 only; N_1 receives no direct input.

Why the EI circuit works without SF: N_1 has no constant drive, so w does not need to overcome a sustained current — it only needs to terminate N_1's plateau, which it does.

Validated Parameters

- $G_{\text{syne}} = 0.063$ nS, $\quad G_{\text{syni}} = 0.05$ nS
- $k_{\text{syn}} = 0.25$, $\quad V_{\text{th}} = -52.0$ mV
- $I_{\text{app}} = 3.0$ pA on N_0 — below 2 pA: no oscillation; above 4 pA: locked above threshold
- $\beta = 0.2$, $\quad \tau_w = 200$ ms

Sweep: `tests/sweep_two_EI.py` updated to remove SF and run a $G_{\text{syne}} \times G_{\text{syni}} \times I_{\text{app}}$ grid (0.01–1.0 nS, I_{app} 2–5 pA). Confirmed oscillations exist in a region around the validated parameters.

Note on sine-wave input: Earlier tests used oscillatory sine-wave input ($f = 0.002$–0.004 cyc/ms). Oscillations at $f = 0.004$ broke down because the input period (250 ms) is comparable to τ_w (200 ms), preventing full w recovery between cycles. The constant-I_{app} test is more informative for assessing intrinsic oscillatory capacity.

Test file: `tests/test_two_EI.py`

4 Single-Segment Network Test

Objective: Apply the EI pacemaker concept to the first segment (17 cells) of the full 102-cell network.

4.1 Network Structure

Each segment contains 17 cells across 9 classes:

Class	Cells	Count	Type
1	0–1	2	Interneuron
2	2	1	Interneuron
3	3	1	Interneuron
4	4	1	Interneuron
5	5–6	2	Motor neuron
6	7–8	2	Motor neuron
7	9–10	2	Motor neuron
8	11–13	3	Dorsal muscle (passive)
9	14–16	3	Ventral muscle (passive)

4.2 Network Directionality

Inter-segment connectivity is asymmetric:

- **Forward (anterior → posterior):** Classes 1, 3, 6 from segment n excite **class 4** in segment $n+1$ — an active oscillator cell.
- **Backward (posterior → anterior):** Class 7 from segment n excites class 4 in segment $n-1$, but classes 1 and 2 project onto **class 8** (passive muscle) — a dead end for signal propagation.

This structural asymmetry biases signal propagation in the anterior-to-posterior direction, consistent with the forward locomotion travelling wave.

4.3 Class-1 Connectivity (Within Segment)

Cell	Excitatory OUT	Excitatory IN	Gap junctions
0	classes 2, 4, 5, 8	class 3	—
1	classes 5, 8, 8	class 3	class 2

Cell 0 is the more influential driver — it connects to class 4, which is the recipient of the inter-segment forward wave.

4.4 EI Pacemaker Design

Since the segment does not self-oscillate with constant drive, two external "head" inhibitory neurons (cells 17–18) were added, forming EI loops with the two class-1 cells:

- Cell 0 ↔ Head neuron 17
- Cell 1 ↔ Head neuron 18

$I_{\mathrm{app}} = 3.0$ pA applied to cells 0 and 1 only. Head neurons receive no direct input and have no SF. Segment internal connectivity retains SF.

Status: EI loops oscillate as intended. Propagation into the rest of the segment through SF-gated synapses is still under investigation.

Test file: `tests/test_single_segment.py`

5 Open Questions

1. Do the EI-driven class-1 oscillations propagate reliably to classes 2–7, or does SF in the segment connectivity attenuate them too strongly?
2. Should SF be replaced by w throughout the segment, or retained as a complementary mechanism?
3. Once the single segment works, how should inter-segment coupling be reintroduced — all six segments at once, or one at a time?
4. Class 4 is missing its inter-segment excitatory drive in the single-segment test (classes 1, 3, 6 from the previous segment). This may need to be added as a synthetic input to get class 4 oscillating reliably.